Technical Document #722: Data Engineering - Comprehensive Overview

Technical Document #722: Data Engineering - Comprehensive Overview

Data lakes store raw data in its native format. They support structured, semi-structured, and unstructured data at any scale.

ETL (Extract, Transform, Load) is a process that extracts data from source systems, transforms it to fit business rules, and loads it into a target data store.

Stream processing handles continuous data streams in real-time. Apache Kafka and Apache Flink are popular stream processing frameworks.

Data warehouses store structured data optimized for analytical queries. They use columnar storage and support complex aggregations.

Partitioning divides data into smaller, more manageable pieces. It improves query performance by reducing the amount of data scanned.

Digital twins are virtual replicas of physical systems. They enable simulation and optimization without risking real-world resources.

Augmented reality overlays digital information on the physical world. It has applications in training, maintenance, and entertainment.

Autonomous vehicles use a combination of sensors, AI, and mapping to navigate without human intervention. Self-driving technology is rapidly advancing.

Key Takeaways:

- Data quality ensures data is accurate, complete, and consistent.
- Data lineage tracks the origin and movement of data through systems.
- Stream processing handles continuous data streams in real-time.